LDA misclassification error

Let $w = \underset{1\times p}{X^T} \underset{p\times p}{\Sigma^{-1}} \underset{p\times 1}{(\mu_1 - \mu_0)}$ $\qquad X\big|_{\text{group } k} \sim N(\mu_k, \Sigma)$

$$R_1(w) = \left\{ w \;\middle|\; w > \tfrac{1}{2}(\mu_1-\mu_0)^T\Sigma^{-1}(\mu_1+\mu_0) - \log\tfrac{\pi_1}{\pi_0} \right\}$$
$$R_0(w) = \left\{ w \;\middle|\; w < \tfrac{1}{2}(\mu_1-\mu_0)^T\Sigma^{-1}(\mu_1+\mu_0) - \log\tfrac{\pi_1}{\pi_0} \right\}$$

Assume $\pi_0 = \pi_1 \Rightarrow \log\frac{\pi_1}{\pi_0} = 0$

$$\mu_{1w} = E(w|Y=1) = \mu_1^T\Sigma^{-1}(\mu_1-\mu_0)$$
$$\mu_{0w} = E(w|Y=0) = \mu_0^T\Sigma^{-1}(\mu_1-\mu_0)$$
$$\sigma_w^2 = Var(w) = (\mu_1-\mu_0)^T\Sigma^{-1}Var(X)\Sigma^{-1}(\mu_1-\mu_0)$$
$$= (\mu_1-\mu_0)^T\Sigma^{-1}(\mu_1-\mu_0) \quad \text{Since } Var(X)=\Sigma$$

using the properties $E(a^TX) = a^T\mu$

$Var(a^TX) = a^TVar(X)a$

So $w\big|_{\text{group } k} \sim N\left(\mu_k^T\Sigma^{-1}(\mu_1-\mu_0),\ \sigma_w^2\right)$

$$\text{Error 1} = P(\text{classify as group 0 but true } Y=1)$$
$$= P(\text{classify as group 0} \mid Y=1)\cdot P(Y=1)$$
$$= \pi_1 P\left(w < \tfrac{1}{2}(\mu_1-\mu_0)^T\Sigma^{-1}(\mu_1+\mu_0)\right)$$
$$= \pi_1 P\left(\frac{w-\mu_{1w}}{\sigma_w} < \frac{\frac{1}{2}(\mu_1-\mu_0)^T\Sigma^{-1}(\mu_1+\mu_0) - \mu_1^T\Sigma^{-1}(\mu_1-\mu_0)}{\sigma_w}\right)$$
$$= \pi_1 P\left(Z < -\tfrac{1}{2}(\mu_1-\mu_0)^T\Sigma^{-1}(\mu_1-\mu_0)/\sigma_w\right)$$
$$= \pi_1 P\left(Z < -\frac{1}{2}\frac{\sigma_w^2}{\sigma_w}\right) = \pi_1 P\left(Z < -\tfrac{1}{2}\sigma_w\right)$$
$$= \pi_1 \Phi\left(-\tfrac{1}{2}\sigma_w\right)$$

$$\text{Error 2} = P(\text{classify as group 1 but true } Y=0)$$
$$= P(\text{classify as group 1} \mid Y=0)\,P(Y=0)$$
$$= \pi_0 P\left(w > \tfrac{1}{2}(\mu_1-\mu_0)^T\Sigma^{-1}(\mu_1+\mu_0)\right)$$
$$= \pi_0 P\left(\frac{w-\mu_{0w}}{\sigma_w} > \frac{\frac{1}{2}(\mu_1-\mu_0)^T\Sigma^{-1}(\mu_1+\mu_0) - \mu_0^T\Sigma^{-1}(\mu_1-\mu_0)}{\sigma_w}\right)$$
$$= \pi_0 P\left(Z \geq \frac{\frac{1}{2}\sigma_w^2}{\sigma_w}\right) = \pi_0 P\left(Z \geq \frac{\sigma_w}{2}\right) = \pi_0 \Phi\left(-\frac{\sigma_w}{2}\right)$$

$$\text{Error 1} + \text{Error 2} = \pi_1 \Phi(-\tfrac{1}{2}\sigma_w) + \pi_0 \Phi(-\tfrac{1}{2}\sigma_w)$$
$$= (\pi_1 + \pi_0)\, \Phi(-\tfrac{1}{2}\sigma_w)$$
$$= \Phi(-\tfrac{1}{2}\sigma_w)$$

Note in Error 1

$$\tfrac{1}{2}(\mu_1 - \mu_0)^T \Sigma^{-1}(\mu_1 + \mu_0) - \mu_1 w$$

$$= \tfrac{1}{2}(\mu_1 - \mu_0)^T \Sigma^{-1}(\mu_1 + \mu_0) - \mu_1^T \Sigma^{-1}(\mu_1 - \mu_0)$$
$$= \tfrac{1}{2}(\mu_1 - \mu_0)^T \Sigma^{-1}(\mu_1 + \mu_0) - (\mu_1 - \mu_0)^T \Sigma^{-1} \mu_1$$

↳ Σ is symmetric

$$= (\mu_1 - \mu_0)^T \Sigma^{-1}\left(\tfrac{1}{2}\mu_1 + \tfrac{1}{2}\mu_0 - \mu_1\right)$$
$$= (\mu_1 - \mu_0)^T \Sigma^{-1}\left(\tfrac{1}{2}\mu_0 - \tfrac{1}{2}\mu_1\right)$$
$$= -\tfrac{1}{2}(\mu_1 - \mu_0)^T \Sigma^{-1}(\mu_1 - \mu_0)$$